

1. Cellular Neuroscience

Goal: Understand the physical and mathematical properties of the neuron.

- **STUDY:**
 - **Chapter 1: Cellular Components** (p. 2–10). Focus on the specialized anatomy of **dendrites, spines, and axons**.
 - **Chapter 2: Entire Chapter** (p. 33–47). This is the most important section for your math requirements.
 - Pay special attention to "**How Ion Movements Produce Electrical Signals**" and the **Nernst/GHK equations**.
 - **Electrical Circuit Model:** While the book discusses this throughout Chapter 2, ensure you understand the neuron as a capacitor (C_m) and resistor (R_m).
- **SKIP:**
 - Chapter 1: *Imaging the Living Human Brain* (p. 22–28) and *Model Organisms* (Box 1A). These are procedural, not cellular.
 - Chapter 2: *Clinical Applications: Anesthesia* (p. 37).

2. Action Potentials (AP)

Goal: Master the Hodgkin-Huxley model and ion kinetics.

- **STUDY:**
 - **Chapter 3: Entire Chapter.** Focus heavily on the **Voltage Clamp Method** (Box 3A) and the reconstruction of the AP.
 - **Chapter 4: * Ion Channels Underlying APs** (p. 65–68).
 - *Voltage-Gated Ion Channels* (p. 73–77).
 - *Active Transporters* (p. 79–82) – specifically the **Na-K Pump**.
- **SKIP:**
 - Chapter 3: *Multiple Sclerosis* (p. 62).
 - Chapter 4: *Thermosensitive and Mechanosensitive Channels* (p. 78).
 - Chapter 4: *Neurological Diseases* (p. 75).

3. Synapses & Plasticity

Goal: Understand chemical signaling, specific neurotransmitters, and the "why" behind memory (plasticity).

- **STUDY:**
 - **Chapter 5:** Focus on *Electrical vs. Chemical Synapses*, *The Role of Calcium* (p. 95), *Quantal Release*, and **Summation/Integration** (p. 108).
 - **Chapter 6:** Focus specifically on **Glutamate** (AMPA/NMDA receptors), **GABA**, and **Biogenic Amines** (Dopamine).

- **Chapter 7:** Focus on *Dendritic Spines* (Box 7B) and how second messengers work.
- **Chapter 8:** Focus on **LTP** (Long-Term Potentiation), **LTD** (Long-Term Depression), and the **Mechanisms Underlying LTP** (p. 179). This links Ca^{2+} dynamics to plasticity.
- **SKIP:**
 - Chapter 5: *The Tripartite Synapse* (Box 5A).
 - Chapter 6: *Acetylcholine, Purines, and Peptides*. (Unless your syllabus specifically asks, stick to Glutamate, GABA, and Dopamine).
 - Chapter 7: *Nuclear Signaling* (p. 162).
 - Chapter 8: *Behavioral Modification in Aplysia* (p. 172). It's a classic experiment, but if you're short on time, the Hippocampal section (p. 176) is more relevant to modern neuro.

Summary of "Must-Know" Equations & Concepts

As you read, ensure you can define/calculate these:

1. **Nernst Equation:** $E_x = \frac{RT}{zF} \ln \frac{[X]_{\text{out}}}{[X]_{\text{in}}}$
2. **GHK Equation:** Essential for determining RMP when multiple ions are involved.
3. **The n , m , h parameters:** m and h for Na^+ (activation/inactivation) and n for K^+ (activation).
4. **Length Constant (λ) and Time Constant (τ):** Found in Chapter 3; vital for understanding AP propagation.